# E-VIGILANCE PLATFORM - QA TEST EXECUTION REPORT

Quality Assurance Certification | TDD, Postman Newman and Cucumber BDD

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|---|---|---|---|
| **Execution Date:** | October 21, 2026 | **Environment:** | **Local QA / Node.js ESM** |
| **System Under Test:** | E-Vigilance Admin Panel v1.0.0 | **Git Branch:** | **test (origin/test)** |
| **Database Cluster:** | MongoDB Atlas Replica Set | **Overall QA Status:** | **100% PASSED (0 FAILURES)** |

**TIER 1: TDD UNIT / INTEG**

**14 / 14 Passed**

node:test runner (100%)

**TIER 2: POSTMAN / NEWMAN**

**20 / 20 Passed**

12 Endpoints Audited (100%)

**TIER 3: CUCUMBER BDD**

**71 / 71 Steps**

12 Living Scenarios (100%)

**Platform Verification Summary:**

All 105 automated checks executed successfully without regression. All RBAC and API quality gates satisfied.

## 1. Tier 1: TDD and Multi-User Security Matrix (14 Tests)

| # | Test Scenario | Test Persona / Payload | Expected | Status |
|---|---|---|---|---|
| 1 | Unauthenticated Guardrail | GET /api/auth/me (No token) | 401 | [PASS] |
| 2 | HQ Admin Authentication | admin@evigilance.com / admin123 | 200 + JWT | [PASS] |
| 3 | Station Officer Authentication | kamal@evigilance.com / password123 | 200 + JWT | [PASS] |
| 4 | RBAC Boundary: Officer -> Users | GET /api/users (Officer token) | 403 | [PASS] |
| 5 | RBAC Boundary: HQ -> Users | GET /api/users (HQ token) | 200 | [PASS] |
| 6 | Master Violation Catalog | GET /api/violation-catalog | 200 | [PASS] |
| 7 | Citizen Report Ingest (M2M) | POST /api/violations/ingest (Key) | 201 | [PASS] |
| 8 | Read Ingested Violation by ID | GET /api/violations/:id | 200 | [PASS] |
| 9 | Case-Insensitive Email Login | ADMIN@EVIGILANCE.COM | 200 | [PASS] |
| 10 | Wrong Password Attempt | admin@evigilance.com / wrong999 | 401 | [PASS] |
| 11 | Non-Existent User Attempt | ghost@police.gov.lk | 401 | [PASS] |
| 12 | Empty Password Validation | admin@evigilance.com / '' | 400 | [PASS] |
| 13 | Inactive Account Security | deactivated_officer@evigilance.com | 401 | [PASS] |
| 14 | Brute-Force Rate Limiting | Excessive login requests in 15m | 429 | [PASS] |

## 2. Tier 2: Postman and Newman API Automation (20 Assertions / 12 Endpoints)

| Endpoint | Method | Assertion Rules Verified | Latency | Status |
|---|---|---|---|---|
| /api/auth/login | POST | Status 200, JWT token returned, role validated as hq | 242ms | [PASS] |
| /api/auth/login | POST | Status 200, officer token issued, variable chaining | 286ms | [PASS] |
| /api/auth/me | GET | Status 200, user profile authenticated with Bearer token | 84ms | [PASS] |
| /api/users | GET | Status 403 Forbidden for station officer accounts | 152ms | [PASS] |
| /api/police-stations | GET | Status 200, extracts active police stations array | 337ms | [PASS] |
| /api/violation-catalog | GET | Status 200, extracts catalog fine amount and clauses | 254ms | [PASS] |
| /api/violations/ingest | POST | Status 201 Created, sets initial status to open | 512ms | [PASS] |
| /api/violations/:id | GET | Status 200, payload matches newly ingested record | 292ms | [PASS] |
| /api/violations/:id/dispatch | POST | Status 200, nearest station geographically assigned | 804ms | [PASS] |
| /api/dispatches/inbox | GET | Status 200, officer receives dispatched case in inbox | 236ms | [PASS] |
| /api/violations/:id/receive | POST | Status 200/403: Enforces cross-station boundary isolation | 915ms | [PASS] |
| /api/violations/:id/update | PATCH | Status 200/403: Audit log appended with officer identity | 1324ms | [PASS] |

## 3. Tier 3: Cucumber BDD Business Acceptance (12 Scenarios / 71 Steps)

| Feature & Scenario | Gherkin Behavioral Specification | Steps | Status |
|---|---|---|---|
| RBAC: Role Matrix | Given user with role <role>, When calling <endpoint>, Then <expected_status> | 21 | [PASS] |
| Security: Deactivation | Given active officer token, When HQ deactivates, Then immediate 401 | 4 | [PASS] |
| Citizen Report Ingestion | When external system posts with X-Ingest-Key, Then plate normalized | 15 | [PASS] |
| Geo Auto-Dispatch | Given GPS coordinates, When auto-dispatch, Then nearest station assigned | 12 | [PASS] |
| Station Inbox Workflow | Officer views station inbox, moves status to in_review with audit trail | 10 | [PASS] |
| Citation and Fine Mapping | When review is verified, Then citation generated matching catalog fine | 9 | [PASS] |

## 4. Audited Security Guardrails and Domain Integrity

**- Brute-Force Rate Limiting (HTTP 429):**

Enforces express-rate-limit on /api/auth/login. Configured to throttle suspicious login attempts (10/15m in production) while enabling seamless high-throughput automated test execution in development/test environments.

**- Station Jurisdiction Boundary Isolation (HTTP 403):**

Strictly guarantees that an officer assigned to Station A cannot review, receive, or modify cases dispatched to Station B. Verified via Newman and BDD test suites.

**- Four-Eyes Verification Principle (HTTP 403):**

Ensures compliance by requiring that the reviewer who moved a violation into review cannot be the same user who verifies and approves the final legal citation.

**OFFICIAL QA QUALITY CERTIFICATION - PASS**

The E-Vigilance Admin Panel has satisfied all test criteria with a 100% pass rate across 105 automated assertions. The system exhibits robust authorization guards, proper audit logging, and consistent API contract adherence.